Insiders and Outsiders in the Rental Market: Entry Rents, Incomes and Housing Supply in Spain

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Abstract

Official rent statistics in Spain average the rents of all leases in force. Between 2015 and 2023 that index rose by 16.5 log points and median income per consumption unit by 31.0, and rents grew less than incomes in every one of the 703 municipalities covered. Households signing a lease faced different prices: in 2024 new leases cost 15.9 log points more than leases in force per euro of cadastral value, and in the Catalan and Aragonese deposit registers rents at signature grew faster than incomes after 2015, by less when later base years are used and not at all in the Basque Country. A stock-flow model with long, indexed leases accounts for the divergence: demand shocks load on rents at signature, and the index of all leases absorbs them only as leases turn over, at 12–19 % a year. Using a shift-share instrument for immigration-driven population growth, I find that entry rents respond more than the rents of all leases: per percentage point of cumulative inflow, 0.83 against 0.22 log points in the baseline and 0.45 against 0.25 with an estimator that separates the main origin groups. How large the response of the entry gap is depends on which origins identify it. Income per consumption unit shows no detectable response, households grow larger, and fewer Spanish nationals move in, while their out-migration shows no detectable change. Supply constraints, tourism and rent regulation appear to act on the same entry margin, but that evidence is imprecise or descriptive. Affordability indicators based on all leases understate the deterioration faced by households entering the rental market.

Keywords Rental housing · Affordability · Entry rents · Housing supply · Immigration · Spain

JEL Classification R21 · R31 · R38 · J61

1 Introduction

The official index of rents in Spain, which averages the rents of all leases declared in income-tax returns, rose by 16.5 log points between 2015 and 2023, while median household income per consumption unit rose by 31.0. In none of the 703 municipalities with an official index did rents outpace incomes. The rents faced by households signing a lease tell a different story. In 2024 new leases cost 15.9 log points more than all leases in force per euro of cadastral value, and the gap is positive in 99.5 % of the 402 municipalities with more than 20,000 inhabitants. In Catalonia, where leases must be deposited with a public agency, the mean rent of new leases rose by 39.0 log points between 2015 and 2023, 11.7 more than incomes; in Zaragoza, rents at signature rose by 37.6 log points for dwellings on the same street, against 14.9 for all leases and 26.1 for incomes.

This paper argues that much of the deterioration of rental affordability in Spain falls on the households that enter the market, explains why in a model of rent resets, and tests the model's predictions. Official statistics average the rents of all leases. But leases last at least five years, existing rents are updated by the consumer price index or by a legal cap, and only about one lease in six ends each year. In such a market, demand shocks load on the rent set when a dwelling is relet. Sitting tenants are insulated; households that move, form or arrive pay the market price. The affordability of insiders and that of outsiders diverge, in a rental-market counterpart of the insider–outsider distinction of labour economics (Lindbeck and Snower 2001).

Section 4 formalises this idea in a stock-flow model of a local rental market with contract duration, indexed updates and rent resets, in which household size, mobility and turnover respond to the same shocks. Because only the households who sign a lease re-optimise, the price elasticity that absorbs a demand shock is scaled down by the turnover rate, and the reset needed to accommodate newcomers is larger the longer leases last. The model delivers five predictions. A demand shock raises entry rents more than the rent of all leases, which absorbs it only at the rate at which leases turn over. Whether the shock worsens affordability depends on the income of those who enter, which income per household can conceal when households grow larger. The response of entry rents is larger where new supply and the reallocation of dwellings from other uses are inelastic. Quantities adjust through the would-be entrants who do not get in rather than through incumbents who leave. And rules that protect incumbents widen the gap between entry rents and the rents of sitting tenants, while caps on entry rents narrow it at the cost of fewer regulated leases.

The empirical analysis uses open administrative data. Tax statistics give, for 2024, the rent, floor area and cadastral value of new leases and of all leases in every municipality above 20,000 inhabitants and in the postal codes of the largest cities, which allows a value-adjusted measure of the entry gap. Deposit registers in Catalonia, Aragón, the Basque Country and the Comunitat Valenciana record rents at signa-

ture; the Aragón register, with 458,392 residential deposits and the street of each dwelling, also reveals how long leases last. The household income atlas gives income per consumption unit and household size by municipality, the household budget survey gives the rent burden of renters, and the statistics of residential changes give moves between municipalities. The demand shock is the immigration-driven inflow of residents predicted by a shift-share instrument based on the settlement of 33 origin groups in 2003 and their later national growth (Saiz 2007; Goldsmith-Pinkham et al. 2020; Borusyak et al. 2022), built from the same annual predictions as in a companion paper on rents and household occupancy (Fernández-Aguilera 2026).

Five results stand out. First, entry rents respond more than the rent of all leases. Per percentage point of cumulative inflow, the implied entry rent rises by 0.83 log points (Anderson-Rubin set [0.25, 2.21]) and the rent of all leases by 0.22 [−0.09, 0.62]; an efficient estimator that uses the main origin groups as separate instruments gives 0.45 and 0.25. The response of the value-adjusted entry gap, a level observed in 2024, is positive in every specification, but its size is not robust: 0.57 [0.11, 1.74] in the baseline, 0.21 with separate instruments and close to zero when only the Colombian component of the instrument is used. In provinces, where indices of new and existing leases measure the change in the gap since 2015 directly, the response is 0.13 (standard error 0.22). Second, income per consumption unit shows no detectable response (0.08, [−0.42, 0.97]), while household size rises by 1.26 [0.69, 2.51] and income per household by 0.89: measured per household, affordability would appear to improve. Third, the in-migration of Spanish nationals from other municipalities falls by 0.54 moves per 100 residents per point of inflow [−2.00, −0.14], while their out-migration shows no detectable change (−0.01, [−0.40, 0.27]). Quantities, like prices, appear to adjust at entry, although the fall in in-migration is also compatible with native avoidance of immigrant destinations. Fourth, the response of the entry gap is larger where more land is undevelopable (0.49 per standard deviation, $p = 0.021$) and on the coast, and the two cannot be separated in these data. Fifth, tourism and regulation appear to act on the same margin. During the collapse of tourism in 2020–21 more long-term leases were signed in Catalan municipalities with many tourist dwellings, although these municipalities were already on different trends. The caps on updates of 2022–2023 account mechanically for at most 58 % of the widening of the national entry gap. And the Catalan cap on new-lease rents in stressed markets was followed by a rise of 1.8 percentage points in the share of exempt seasonal leases.

Three limits qualify these results and are discussed where they arise. The entry gap is observed in levels only in 2024, so its response is identified from a cross-section and translates into a response of entry-rent growth only under an assumption on the gap in 2015. The instrument draws its identifying variation from the 2003 settlement of a few origin groups, whose shares are not balanced on predetermined characteristics and whose origin-specific estimates differ. And the incomes of the households

who sign new leases are not observed, so the affordability of entrants is inferred from local averages.

The paper contributes to three literatures. It adds to the economics of rental affordability (Quigley and Raphael 2004; Khametshin et al. 2024) a distinction between insiders and outsiders, a model of why they diverge in markets with long indexed leases, and evidence that they do. It contributes to the measurement of rents by showing, with tax data and deposit registers, how much indices of all leases (Genesove 2003; Ambrose et al. 2015; Crone et al. 2010; Adams et al. 2024) understate what entrants pay in a market with long statutory leases, and by constructing a value-adjusted entry gap. And it links the literatures on local demand shocks and supply constraints (Saiz 2010; Glaeser and Gyourko 2018; Hilber and Vermeulen 2016) and on rent regulation (Arnott 1995; Autor et al. 2014; Diamond et al. 2019; Jofre-Monseny et al. 2023) through the margin at which adjustment takes place: in a market that protects incumbents, prices and quantities appear to adjust at entry.

2 Related literature

Rental affordability is usually measured as the ratio of rent to the income of renters (Quigley and Raphael 2004). For Spain, Khametshin et al. (2024) document the rise in rental demand since 2015 among young and foreign-born households and in large urban and tourist areas, and the high rent burden of low-income tenants, and describe the quantification of its causes as an open question. That rents paid by sitting tenants differ from market rents is well known. Tenants who stay longer pay less (Goodman and Kawai 1985; Ambrose et al. 2015), existing rents are rigid in nominal terms (Genesove 2003), and official rent indices have understated rent growth (Crone et al. 2010). Indices of rents on new leases lead indices of all leases (Adams et al. 2024). These studies concern markets with short or open-ended leases; Spain combines statutory minimum durations of five to seven years with indexed updates and, recently, caps, which makes the wedge larger and slower to close.

Rent regulation turns the wedge into a policy instrument. Tenancy rent control, which limits increases within a tenancy but leaves rents free between tenancies, protects incumbents and encourages landlords to front-load entry rents (Arnott 1995; Basu and Emerson 2000). Rent control lowers tenant mobility and misallocates dwellings (Glaeser and Luttmer 2003; Sims 2007; Diamond et al. 2019), and it reduces the supply of rental housing to newcomers (Autor et al. 2014; Diamond et al. 2019; Kholodilin 2024). In Catalonia, rent caps lowered rents and the number of leases signed (Jofre-Monseny et al. 2023; Monras and Montalvo 2023; Pérez García 2026).

Local demand shocks are capitalised into rents where supply is inelastic (Roback 1982; Saiz 2010; Glaeser and Gyourko 2018; Hilber and Vermeulen 2016; Hsieh and Moretti 2019). Immigration is a standard source of such shocks: Saiz (2007) finds that an inflow of 1 % of a city's population raises rents by about 1 %, Sá (2015) finds that immigration lowered house prices in the United Kingdom as natives

moved out, and Gonzalez and Ortega (2013) and Sanchis-Guarner (2023) study house prices in Spain. Whether natives leave or avoid immigrant destinations is itself a margin of adjustment (Card 2001; Saiz and Wachter 2011). Shift-share designs identify such effects either from the exogeneity of the shocks (Borusyak et al. 2022; Adão et al. 2019; Borusyak and Hull 2023) or from that of the initial shares (Goldsmith-Pinkham et al. 2020), and immigration instruments based on past settlement can confound current and past inflows (Jaeger et al. 2018). Short-term rentals reallocate dwellings from residents to visitors (Barron et al. 2021); in Barcelona they raised rents (Garcia-López et al. 2020), and in Lisbon the collapse of tourism in 2020 lowered rents and increased the long-term rental supply (Batalha et al. 2022). None of these strands separates the rents paid by insiders from those faced by entrants, which is the focus of this paper.

3 Institutions, data and measurement

3.1 Leases, updates and regulation

Since March 2019, residential leases in Spain last at least five years when the landlord is a natural person and seven when it is a company. Annual updates of existing leases are usually tied to the consumer price index. From the end of March 2022 to the end of 2023, updates were limited to the change in the competitiveness guarantee index, itself capped at 2 %; landlords who were not large holders could agree a different update with the tenant, and the limit applied in the absence of agreement. Updates were capped at 3 % in 2024, and since 2025 they are limited by a reference index published by INE. Rents are set freely at the start of a new lease except in stressed (tensioned) market areas declared under the 2023 Housing Act, where they are capped by the rent of the previous lease and, for some landlords, by a reference index. Catalonia applied this cap in 140 municipalities from 16 March 2024 and in 131 more from October 2024. Seasonal leases, which the Urban Leases Act treats as a use other than housing, were exempt from the cap throughout the period studied; their regime was changed in Catalonia in 2026 and nationally in October 2026, after the end of the data.

3.2 Rents of all leases and rents at entry

All leases. The INE rent index (IPVA) averages the rents of all leases in force declared in income-tax returns, excluding the Basque Country and Navarre, by municipality above 10,000 inhabitants. Since 2021 INE also publishes, with base 2015 = 100, separate indices for new and existing leases by province, with a methodological break in that year; the implied weight of new leases was 17.6 % in 2021 and 12.3 % in 2024.

The entry gap. The tax agency publishes, for 2024, the average rent, floor area and cadastral reference value of all leases declared and of leases that started in the year, by municipality above 20,000 inhabitants

and by postal code in the largest cities. The main outcome is the value-adjusted entry gap,

$$G_i^{VR} = \ln\Big(\frac{\text{rent}_i^{new}}{\text{value}_i^{new}}\Big) - \ln\Big(\frac{\text{rent}_i^{all}}{\text{value}_i^{all}}\Big),$$

the log ratio of gross yields on new and on all leases, where value is the mean cadastral reference value of the dwellings. In Spain as a whole new leases averaged 816 euros per month against 729 for all leases, on dwellings of lower value (145,067 against 154,022 euros), so the gross yield was 6.8 % against 5.7 %. Cadastral values adjust for location and size but not for the condition of the dwelling, so I call the measure value-adjusted rather than quality-adjusted; versions per m^2, raw and with value elasticities below one are reported. Four features of the measure matter for interpretation. It covers landlords who are natural persons and declare in personal income tax, not companies. Leases that started in the year include new contracts with the same tenant, which dilute the gap. The comparison group, all leases, contains the new ones: with a share of new leases of 12.3 %, a gap of 15.9 log points relative to all leases corresponds to about 18.4 relative to existing leases. And the 2023 Housing Act changed the income-tax reductions on rental income for leases signed after its entry into force, so new and old leases faced different tax incentives in 2024, which may affect declared rents.

Deposit registers. Leases must be accompanied by a deposit lodged with a regional agency. Registers record rents at signature in Catalonia (mean rent by municipality and quarter, since 2005), Aragón (one record per deposit with rent, street and whether the deposit has been returned, since 1996), the Basque Country (rent per m^2 built, municipality and year, since 2016) and the Comunitat Valenciana (deposit amount, since 2020). In Aragón I match 99.95 % of residential deposits to municipalities and build an index of entry rents from a regression of log rent on year effects and street fixed effects within each municipality, which holds location constant.

3.3 Incomes, households, moves and the demand shock

The INE household income atlas (2015–2023) gives median net income per consumption unit, income per person and per household, and average household size by municipality. The INE household budget survey (2016–2024) gives the rents, incomes and floor area of households renting at market price (23,274 household-years); because exact income is reported for only 42 % of renters in 2024, I use the midpoint of the reported income interval in every year. The INE statistics of migrations and changes of residence give moves between municipalities by nationality (2021–2024); naturalised residents count as Spanish nationals.

The demand shock is the cumulative net inflow of foreign-born residents between 1 January 2015 and 1 January 2025 relative to 2015 population, from the municipal register (to 2021) and the annual census (from 2022), using changes within each source. Supply constraints are the shares of land within

10 km of each municipality that are sea or have a slope above 15 %, computed from the Copernicus 90-metre digital elevation model in the spirit of Saiz (2010); cadastral dwelling counts measure the supply response. Tourism is measured by tourist dwellings as a share of all dwellings in August 2020, the first month of the INE experimental statistics. Regulation is measured by seasonal leases by municipality and quarter in Catalonia (2023–2025) and by inflation and the legal caps on updates. Tenure exposure is the appraised housing value relative to income by municipality (Ministry of Housing). All data are public and are downloaded by the replication code.

4 A model of insiders, entrants and rent resets

The model describes a local rental market i in which leases are long, existing rents follow an update rule and a market price applies only when a dwelling is relet. It has three blocks: an accounting block that links the rent of all leases to entry rents through turnover; a market block that determines entry rents and household size; and a behavioural block in which turnover and migration respond to the gap between the two rents. Lower-case letters denote logs, t indexes years, and the market subscript i is dropped where it causes no confusion.

4.1 Leases, resets and the index of all leases

In year t a share λ_t of leases ends and the dwelling is relet at the market-clearing entry rent R_t^*; the other leases are updated by $u_t = \min(\pi_t, \bar{u}_t)$, inflation or a legal cap. Treating the log of the average rent as the average of log rents, a first-order approximation, the log rent of all leases $\bar{r}_t$ satisfies

$$\bar{r}_t = \lambda_t \, r_t^* + (1 - \lambda_t)\,(\bar{r}_{t-1} + u_t), \tag{1}$$

and the entry gap $G_t \equiv r_t^* - \bar{r}_t$ evolves as

$$G_t = (1 - \lambda_t)\left[G_{t-1} + \Delta r_t^* - u_t\right]. \tag{2}$$

The gap widens, $G_t > G_{t-1}$, when $\Delta r_t^* - u_t > \lambda_t G_{t-1}/(1-\lambda_t)$: entry rents must outgrow updates by more than is needed to sustain the existing gap. As in models of staggered price setting with a constant hazard of resetting (Calvo 1983), the rent of all leases is a distributed lag of past entry rents.

Three implications follow. First, if entry rents grow at a constant rate g and updates at u, the gap converges to

$$G^* = \frac{1-\lambda}{\lambda}\,(g - u). \tag{3}$$

With the turnover measured in section 5, λ between 0.12 and 0.19, the factor $(1-\lambda)/\lambda$ is between 4.2 and 7.1: entry rents that outgrow updates by two points a year sustain a gap of 8.4 to 14.2 log points.

Second, a permanent shock to entry rents passes to the index of all leases at rate λ: the index absorbs a share $1-(1-\lambda)^k$ of the shock after k years, 12–19 % after one year and 41–57 % after four. Third, iterating (2),

$$G_t = \sum_{j=0}^{J-1} \Big[\prod_{s=0}^{j} (1-\lambda_{t-s}) \Big] (\Delta r^*_{t-j} - u_{t-j}) + \Big[\prod_{s=0}^{J-1} (1-\lambda_{t-s}) \Big] G_{t-J} : \tag{4}$$

the gap in a given year reflects mainly recent excess growth of entry rents, and the gap of nine years earlier enters with a weight $(1-\lambda)^9$ of 15–31 %. Longer minimum durations lower λ and caps on updates lower u; both widen the gap for a given path of entry rents.

4.2 Entry rents, incomes and household size

Entry rents clear the market for the dwellings that change hands. Let H be the stock of dwellings let as main residences and N the population of the market. Each year the households whose leases end, a share λ, and the net new population choose how much housing to rent at the entry rent, while sitting tenants keep the space they contracted. An entrant consumes $q = A\,(Y/S)^{\alpha}\,(R^*)^{-\eta}$ units of housing per person, where Y/S is income per consumption unit, S an equivalence scale that rises with household size, and η the price elasticity of space per person: a high η means that higher rents are absorbed by sharing dwellings. Supply comes from new construction, with elasticity σ_i with respect to the entry rent, and from dwellings reallocated from tourist use or vacancy, with elasticity ε_i, both relative to the stock. Starting from a stationary equilibrium with $H = Nq$, market clearing after a population shock dN requires $(1-\lambda)Nq + (\lambda N + dN)(q+dq) = H\,[1 + (\sigma_i + \varepsilon_i)\,dr^*]$, and to first order

$$d\,r^* = \frac{d\ln N + \lambda\,\alpha\,(d\ln Y - d\ln S)}{\lambda\,\eta + \sigma_i + \varepsilon_i}. \tag{5}$$

Because only entrants re-optimise, the price elasticity of demand that absorbs a population shock is $\lambda\eta$ rather than η: the lower the turnover, the larger the reset needed to make room for newcomers. On impact the entry gap rises by $(1-\lambda)\,dr^*$ and the rent of all leases by $\lambda\,dr^*$. Both terms imply that the entry gap responds more to demand where turnover is lower, a prediction that separates this model from one in which all rents adjust; the data used here cannot test it, because turnover is observed only in Aragón.

The affordability of entrants, $A^E = r^* - \ln(Y/S)$, changes by

$$dA^E = d\,r^* - \big(d\ln Y - d\ln S\big), \tag{6}$$

while that of insiders, $A^I = \bar{r} - \ln(Y/S)$, changes on impact by $\lambda\,d\,r^* - (d\ln Y - d\ln S)$. When the shock comes from arrivals with incomes below the local mean, $d\ln Y - d\ln S$ is small or negative. Entrants absorb part of the shock by consuming less space per person, which shows up as larger households, and income per household rises with household size even if income per consumption unit does not. Average

household size can also rise through composition, because arriving households are larger, without any household sharing more; the model does not separate the two. Tourism enters through ε_i. Tourist demand withdraws dwellings from long-term letting and a collapse of tourist demand returns them, but the stock that can be returned is small where tourist lets are rare, so that ε_i is close to zero below some share of dwellings in tourist use.

4.3 Turnover, lock-in and who adjusts

Turnover responds to the gap through two opposite forces. A sitting tenant who moves within the market gives up a rent that is G below the market, so tenant mobility falls with the gap, as with the length-of-residence discounts of Goodman and Kawai (1985) and the lock-in effects of rent control (Glaeser and Luttmer 2003; Sims 2007; Diamond et al. 2019). A landlord, conversely, gains G by ending a lease at expiry and reletting, so landlord-initiated turnover rises with the gap where the law allows it. Writing

$$\lambda_t = \max\{\lambda_{\min},\ \lambda_0 - \kappa G_t\}, \tag{7}$$

the sign of κ is an empirical matter. The floor $\lambda_{\min} > 0$ reflects moves for reasons unrelated to rents, such as jobs, household formation and dissolution, or death. It guarantees a finite steady state: with $\kappa > 0$, substituting (7) into (3) makes the gap feed on itself, and without the floor turnover could collapse and the gap grow without bound. With $\kappa > 0$, a shock that widens the gap lowers turnover, makes the gap more persistent and reduces the number of dwellings that come back to the market each year: the flow of vacancies shrinks when entry demand rises.

The same logic governs migration. Would-be entrants from other municipalities compare entry rents across markets, so in-migration to market i changes by $d\ln M_i = -\zeta\, dr_i^*$, where $\zeta \geq 0$ is the location elasticity of entrants. Sitting tenants face only the update and the gap they would forgo by moving, so their out-migration changes by $d\ln O_i = \xi\,\lambda\, dr_i^* - \kappa_O\, dG_i$, with $\xi \geq 0$ and $\kappa_O \geq 0$. A demand shock therefore lowers in-migration in proportion to $\zeta\, dr^*$ and raises the out-migration of incumbents by at most $\xi\,\lambda\, dr^*$, and by less if lock-in operates. If all rents adjusted ($\lambda = 1$), incumbents would face the full increase and their out-migration would rise by $\xi\, dr^*$.

4.4 Regulation

Equation (2) gives the mechanical effect of a cap on updates: it lowers u_t and widens the gap one for one for the leases updated while it binds. A forward-looking landlord adds a second effect (Basu and Emerson 2000). Suppose that the landlord sets the entry rent so that the expected discounted real rent over the life of the lease is unchanged, and expects updates to fall short of inflation by $\pi - \bar{u}$ in each year of the lease. The entry rent then rises by approximately $(\pi - \bar{u})\,\bar{k}$, where $\bar{k} = \rho(1-\lambda)/[1-\rho(1-\lambda)]$

is the discounted expected time elapsed within a lease and ρ the discount factor. With λ between 0.12 and 0.19 and $\rho = 0.96$, $\bar{k}$ is between 3.5 and 5.3 years. A permanent and anticipated cap therefore raises entry rents through front-loading; an unexpected and temporary cap, like the Spanish one of 2022–2024, mainly widens the gap mechanically. A cap on entry rents lowers r^* in the regulated segment but also lowers the return on letting there, so owners who can move dwellings to unregulated uses, such as seasonal lets, do so, and the number of regulated leases falls.

4.5 Predictions

P1 (entry loading). A demand shock raises entry rents by more than the rent of all leases, which absorbs it at the rate λ, so that the entry gap widens. The response of the gap is larger where turnover is lower.

P2 (income measurement). A shock that raises household size raises income per household without raising income per consumption unit. Affordability measured per household can improve while the affordability of entrants deteriorates.

P3 (amplification). The response of entry rents is larger where new supply (σ_i) and reallocation (ε_i) are less elastic: in land-constrained markets and where few dwellings can be moved into long-term letting from other uses. Reallocation from tourist use is non-linear in the share of tourist lets.

P4 (who adjusts). A demand shock lowers the in-migration of would-be entrants and changes the out-migration of incumbents little. If tenant lock-in dominates ($\kappa > 0$), turnover falls as the gap widens.

P5 (regulation). Caps on updates widen the gap; caps on entry rents narrow it and reduce the number of regulated leases, shifting lets towards unregulated uses.

4.6 Incidence and scope

The model also says who bears the cost of a demand shock. With λ the share of dwellings relet in a year, the extra rent paid in the year of the shock is $\lambda\, dr^*$ per dwelling, all of it by entrants; sitting tenants pay only the update. Until their leases end, the landlords of sitting tenants forgo the market rent, and the landlords who relet capture it. Affordability statistics based on all leases or on all renters average the two groups with weights λ and $1 - \lambda$, and so register only a share λ, 12–19 %, of the change faced by entrants in the year of the shock.

The model is deliberately partial. It treats dwellings as homogeneous, ignores how landlords select tenants, takes construction and reallocation elasticities as given rather than responding to the gap, and does not model the choice between renting and owning. It is a framework for organising the evidence, not a structural model to be estimated.

5 Insiders and entrants

Table 1 and figure 1 set out the central facts. For all leases, rents grew by 16.5 log points between 2015 and 2023 and income per consumption unit by 31.0: the rent-to-income ratio of sitting tenants fell by 14.4 log points on average, by 19.5 in municipalities of 10,000–20,000 inhabitants and by 7.9 in the largest cities, and it fell in every municipality. The household budget survey points the same way for renters as a whole: their median rent-to-income ratio was 27.6 % in 2016 and 25.4 % in 2024, and the share paying more than 40 % of income fell from 26 % to 18 %.

New leases tell a different story in three of the four regions with registers. In Catalonia, rents on new leases rose by 39.0 log points against 27.3 for incomes, and they outpaced incomes in 77 % of municipalities; the difference between entry-rent and income growth increases with the size of the municipality (figure 1b). In Aragón, where the street fixed effects hold the location of the dwelling constant, rents at signature rose by 36.1 log points against 26.5 for incomes and 15.1 for all leases. They outpaced incomes in 77 % of the 43 municipalities with enough leases, and outpaced the index of all leases in each of the 14 that have one. In the Comunitat Valenciana, deposits on new leases rose by 22.9 log points between 2020 and 2023, against 15.9 for incomes. In the Basque Country, however, rents per m^2 on new leases rose by 21.2 log points between 2016 and 2023, less than incomes (26.4), and they outpaced incomes in only 10 % of municipalities.

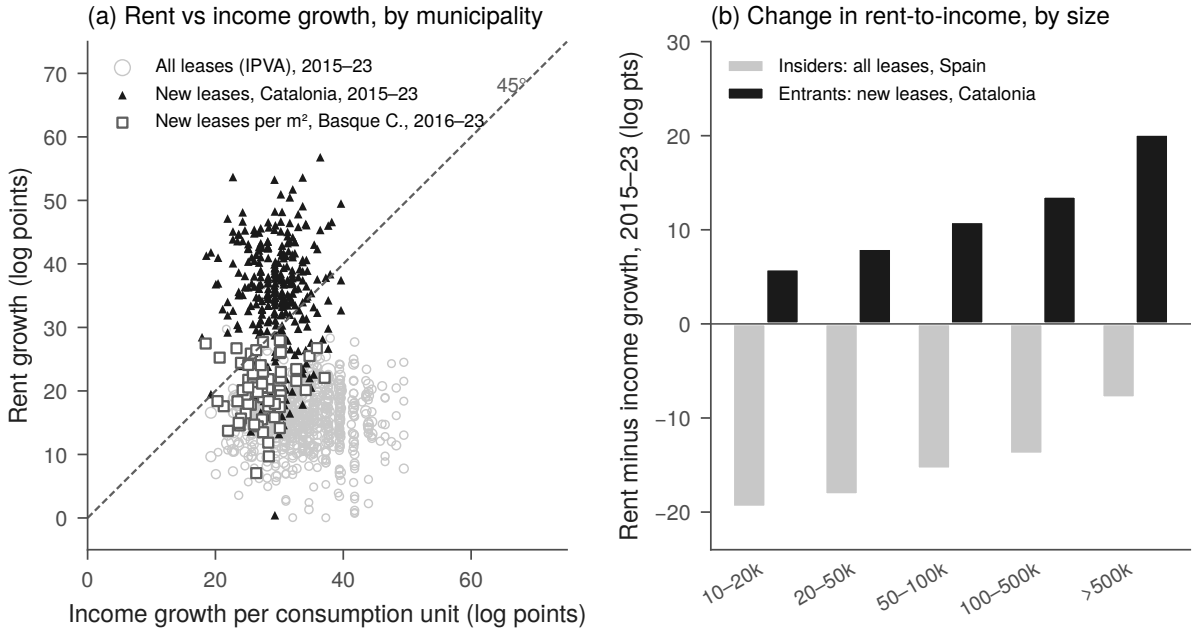


Fig. 1 Rents of insiders and entrants against incomes

Notes: panel (a), growth of rents and of median income per consumption unit by municipality: index of all leases (IPVA, 2015–2023), mean rent of new leases in Catalonia (2015–2023, municipalities with at least 30 leases in 2015) and rent per m^2 of new leases in the Basque Country (2016–2023). Panel (b), population-weighted difference between rent and income growth by size of municipality. Sources: INE, Generalitat de Catalunya, Gobierno Vasco.

These comparisons start in 2015, close to the trough of the rental cycle, and the base year matters.

Table 1 Rents of insiders and entrants against incomes

Rent measure	Period	Units	Rent growth	Income growth	Difference	Share with rent > income
			(log points, weighted)			
All leases (IPVA), Spain	2015–2023	703 municipalities	16.5	31.0	−14.4	0 %
New leases (deposits), Catalonia	2015–2023	289 municipalities	39.0	27.3	11.7	77 %
New leases per m^2 (deposits), Basque Country	2016–2023	67 municipalities	21.2	26.4	−5.2	10 %
New leases, street-adjusted index (deposits), Aragón	2015–2023	43 municipalities	36.1	26.5	9.5	77 %
Deposits on new leases, Comunitat Valenciana	2020–2023	104 municipalities	22.9	15.9	7.0	–
Entry gap: new leases relative to leases in force						
			New leases	Existing or all	Gap	Share with gap > 0
INE index, new vs existing, Spain	2021–2024	national	17.6	7.2	10.4	
AEAT, new vs all leases, per euro of cadastral value	2024	402 municipalities			15.9	99.5 %[a]

Notes: income is the median net income per consumption unit of the municipality (INE household income atlas). Means weighted by 2015 population. IPVA: index of all leases in force (tax returns). Deposit registers record leases at signature: mean rent in Catalonia (contracts of more than one year), rent per m^2 built in the Basque Country (free-market collective main residences), an index of rents per contract with street fixed effects in Aragón, deposit amount in the Comunitat Valenciana (one month of rent by law; partial coverage). AEAT: tax statistics of dwellings declared in 2024, municipalities above 20,000 inhabitants; the entry gap is $\ln(\text{rent}_{new}/\text{value}_{new}) - \ln(\text{rent}_{all}/\text{value}_{all})$. [a] Share of municipalities with a positive gap.

In the Catalan register the mean rent of new leases in 2015 was 19.7 log points below its 2008 level and rose by 40.5 by 2023, so part of the growth after 2015 is a recovery. Measured from 2018, rents at signature grew by 1.7 log points more than incomes in Catalonia and 4.0 in Aragón, against 11.7 and 9.5 from 2015 (table A2, panel B). The divergence between entry rents and incomes is concentrated in the years of recovery, especially in Catalonia; the divergence between entry rents and the rents of sitting tenants, by contrast, widened after 2021. Nor is the Basque exception by itself evidence against the mechanism: within the Basque Country, the excess of entry-rent growth over income growth was larger in municipalities with larger foreign-born inflows (0.65 per point of inflow, standard error 0.28; table A2, panel C).

The entry gap is widening nationally. Relative to 2015, the INE index of new leases exceeded the index of existing leases by 2.3 log points in 2021 and by 12.7 in 2024, and the gap widened in every province (figure 2a). In levels, the tax data show a value-adjusted gap of 15.9 log points in 2024, with medians between 14.9 and 16.1 log points across size classes (figure 2b). New leases are signed on dwellings of lower value and similar size, so the raw gap (13.9) is smaller than the value-adjusted one; with value elasticities of 0.5 and 0.7 the mean gap is 14.9 and 15.3 log points (table A2, panel A). Within the largest cities the gap is positive in 99.4 % of postal codes (mean 17.1 log points), 65 % of its variance is within cities, and it is unrelated to the rent level of the neighbourhood (slope −0.011, standard error 0.015). The entry gap is a feature of the market rather than of particular neighbourhoods.

How fast leases turn over determines how slowly the index of all leases follows the market. The Aragón register shows, for each cohort of deposits, the share not yet returned at the date of download. A survival curve with a constant hazard and a small share of deposits never returned (3.5 %) fits these shares closely (figure 3b) and implies that 17.4 % of leases end each year (14.9 % in Zaragoza), with a median duration of 3.8 years. Measuring the age of each cohort from the start of the year or to the

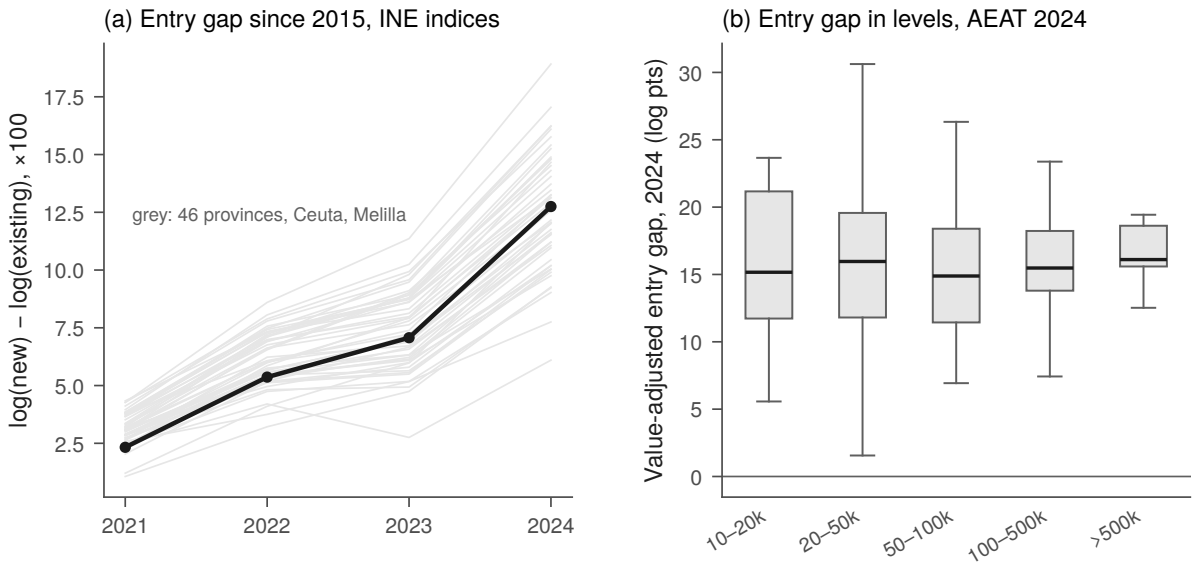


Fig. 2 The entry gap over time and in levels

Notes: panel (a), $100 \times \ln(\text{index}_{new}/\text{index}_{existing})$, both indices with base 2015 = 100 (INE table 59005), Spain and the 46 provinces of the common tax regime plus Ceuta and Melilla. Panel (b), value-adjusted entry gap in 2024 (AEAT) by size of municipality: median, interquartile range and 1.5 interquartile-range whiskers.

start of 2026 instead of from mid-year moves the rate between 14.9 % and 19.2 %. Exit rates are similar for cohorts signed before and after the 2019 reform that lengthened minimum durations (17.4 % and 17.4 %). Two caveats apply: a deposit can remain unreturned after the lease ends if no one claims it, which biases the exit rate downwards, and the register mixes leases signed under different legal regimes. The weight of new leases implied by the INE indices, 17.6 % in 2021 and 12.3 % in 2024, lies in the same range. Its fall as the gap widened is what tenant lock-in would produce, but it coincides with the longer minimum durations of 2019 and with the update caps, and is not a test of lock-in. With λ between 0.12 and 0.19, equation (3) implies that entry rents growing two points a year faster than updates sustain a gap of 8.4 to 14.2 log points, the order of magnitude observed in 2024.

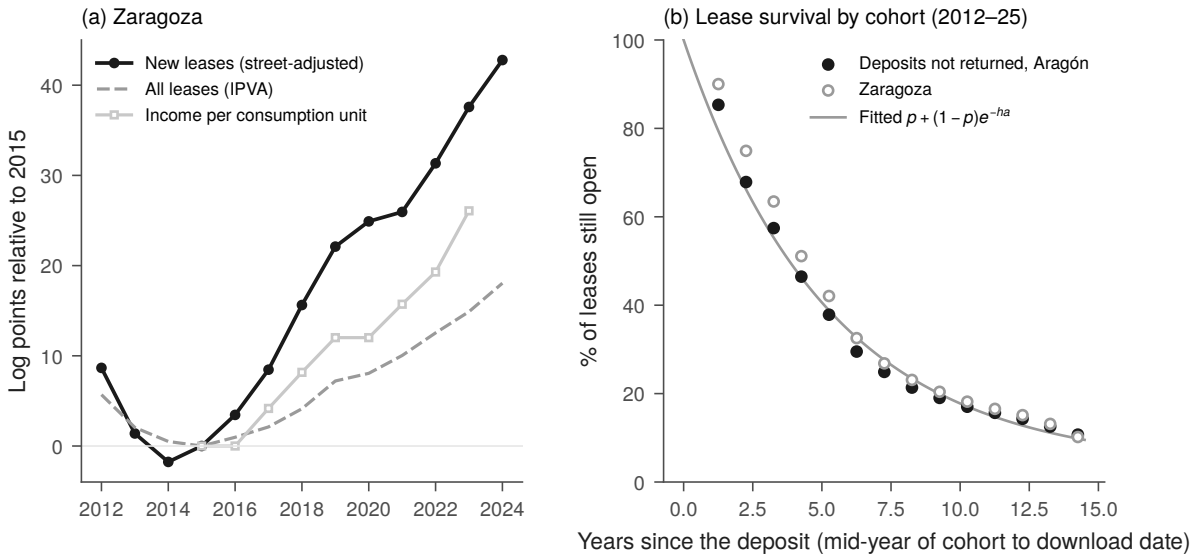


Fig. 3 Entry rents and lease turnover in the Aragón deposit register

Notes: panel (a), Zaragoza: index of rents of new leases with street fixed effects (Aragón deposit register), IPVA index of all leases and median income per consumption unit, log points relative to 2015. Panel (b), share of each cohort of residential deposits (2012–2025) not returned at the date of download, Aragón and Zaragoza, by years since the middle of the cohort year, and fitted survival curve $S(a) = p + (1-p)e^{-ha}$ for Aragón. Sources: Gobierno de Aragón, INE.

6 Empirical strategy

For municipality i in province p,

$$y_i = \beta\, x_i + \boldsymbol{w}_i'\boldsymbol{\gamma} + \delta_p + e_i, \tag{8}$$

where x_i is the cumulative foreign-born inflow relative to 2015 population, in percentage points, and y_i is, in turn, the entry gap in 2024, the growth of the rent of all leases, implied entry-rent growth (the sum of the two), income per consumption unit, income per person and per household, household size, population, the dwelling stock and moves between municipalities. For outcomes measured to 2023 the inflow ends on 1 January 2024. Affordability outcomes are rent growth minus growth of income per consumption unit, for insiders (all leases) and for entrants (implied entry rents). Coefficients are in log points per percentage point of inflow.

Instrument. x_i is instrumented with

$$z_i = \sum_o \ell_{io}\, \frac{\Delta F_o - \Delta F_{op}}{P_{i,2015}}, \tag{9}$$

where $\ell_{io} = F_{io,2003}/F_{o,2003}$ is municipality i's share of the 2003 national stock of residents born in origin o (33 origins), $\Delta F_o - \Delta F_{op}$ is the national change in that stock over the same years excluding province p, and $P_{i,2015}$ is the 2015 population. The instrument is the sum over 2015–2024 of the annual leave-own-province-out predictions of the companion paper (Fernández-Aguilera 2026). The two papers overlap: the companion paper uses the 699 municipalities of the official index to study the rent of all leases and

household occupancy, while this paper adds the entry gap, incomes per consumption unit, moves, the deposit registers and the model; its estimates for the rent of all leases are reported for comparison. Controls are predetermined characteristics from the 2011 census and 2012 firm counts, and the specification includes province fixed effects. Regressions are weighted by 2015 population and standard errors are clustered by province with a small-sample correction; the entry-gap sample has 42 provinces. Because the first stage is moderate, I report Anderson-Rubin sets and their wild cluster restricted bootstrap version throughout.

Estimand. With heterogeneous effects, two-stage least squares identifies a weighted average of municipal effects, with weights that increase with population and with the strength of the instrument's prediction of the inflow. In practice these are municipalities where the 2003 settlement of a few origin groups predicted later inflows. The estimates are a population-weighted local effect of immigration-driven population growth, not the effect of any population shock.

Identification. The instrument can be justified by the exogeneity of the origin shocks (Borusyak et al. 2022) or by that of the 2003 settlement shares (Goldsmith-Pinkham et al. 2020). In a long difference there is a single realisation of 33 origin shocks, and the shocks are heavy-tailed because some origins had few residents in 2003. Section 8 shows that shock-based inference (Adão et al. 2019; Borusyak and Hull 2023) is uninformative in this setting. The identifying assumption is therefore, in practice, that the 2003 shares of the origins that carry the instrument (Colombia, Peru and Romania, with 84 % of the Rotemberg weight) are unrelated to later changes in the outcomes, conditional on controls. These shares are not balanced on predetermined characteristics (10 of 18 balance coefficients are significant at 5 %; table A1). I assess the assumption with the sum-of-shares control of Borusyak et al. (2022), a control for the 2003–2014 inflow (Jaeger et al. 2018), separate instruments by origin with an over-identification test, the omission of each main origin, a pre-period falsification test and an annual panel in which shocks vary by year.

Levels and changes. The entry gap is observed in 2024 only. By (4), its response to the instrument measures the response of entry-rent growth relative to updates over 2015–2024 if the 2015 gap is unrelated to the instrument conditional on controls. Otherwise the estimate is biased by $(1-\lambda)^9$ times the coefficient of the 2015 gap on the instrument, a factor of 15–31 %. The implied entry rent, the sum of the responses of the rent of all leases and of the gap, measures the response of entry rents since 2015 under the same assumption. I assess it in three ways: with the pre-period test on all leases (2012–2015); with the provincial INE indices of new and existing leases, which measure the change in the gap since 2015 directly; and by separating earlier from recent inflows, since equation (4) implies that the 2024 gap should respond mainly to recent ones.

Supply constraints. Prediction P3 is tested by interacting the inflow with standardised measures

of geographic constraints and instrumenting the interaction with z_i times the constraint. Geographic constraints are predetermined and, unlike density or construction, do not reflect past demand; but they are correlated with being on the coast, which may matter for other reasons, so I also interact the inflow with a coastal indicator.

7 Results

7.1 Where a demand shock goes

Table 2 and figure 4 report the effects of the demand shock. A cumulative inflow of one percentage point of the population raises the value-adjusted entry gap in 2024 by 0.57 log points (standard error 0.30; Anderson-Rubin set [0.11, 1.74]; wild-bootstrap set [0.10, 1.54]; first-stage F 11.0), the gap per m^2 by 0.50 and the raw gap by 0.71. The rent of all leases rises by 0.22 [−0.09, 0.62], so the implied response of entry rents is 0.83 [0.25, 2.21], several times that of the rent of all leases, as prediction P1 implies. Section 8 shows that the size of the response of the gap depends on how the origin groups are weighted, while the ordering of entry rents above the rent of all leases does not.

The model gives a benchmark for the relative size of the two responses. Given the national timing of the inflow, which was concentrated in 2018–2019 and 2022–2024, and λ between 0.12 and 0.19, the index of all leases should have absorbed 36–50 % of the response of entry rents by 2024. The baseline estimates imply 27 % and the over-identified estimates of section 8 54 %, on either side of that range; the confidence sets are too wide to tell either apart from the model's range or from each other.

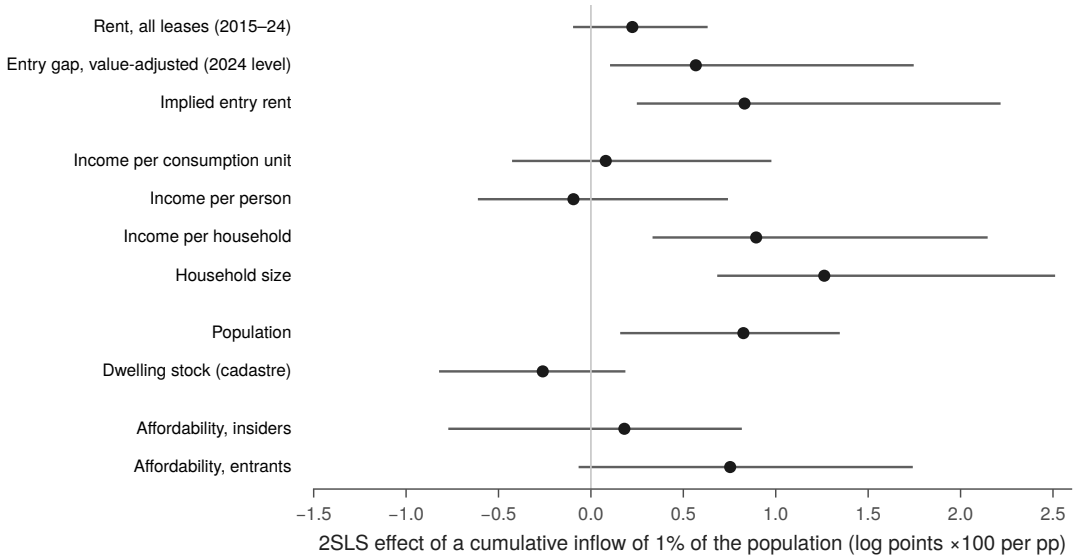


Fig. 4 Where a local demand shock goes

Notes: 2SLS coefficients of table 2 (log points per percentage point of inflow) and Anderson-Rubin 95 % sets; arrows mark sets that extend beyond the axis.

The shock raises population by 0.82 log points per point of inflow [0.17, 1.34]: most of the inflow adds to the population. Section 7.2 shows that fewer Spanish nationals moved in during 2021–2024,

Table 2 Where a local demand shock goes: rents, entry rents, incomes and households

Outcome	2SLS	SE	AR 95 % set	WCR-AR 95 % set	First-stage F	N	Mean
A. Rents							
Rent, all leases, 2015–2024	0.22	(0.16)	[−0.09, 0.62]	[−0.06, 0.64]	21.0	695	0.200
Entry gap, value-adjusted, 2024 (level)	0.57	(0.30)	[0.11, 1.74]	[0.10, 1.54]	11.0	402	0.159
Entry gap, per m^2, 2024 (level)	0.50	(0.32)	[0.06, 1.90]	–	11.0	402	0.163
Entry gap, raw, 2024 (level)	0.71	(0.27)	[0.30, 1.73]	–	11.0	402	0.139
Implied entry rent (rent of all leases + value-adjusted gap)[a]	0.83	(0.37)	[0.25, 2.21]	[0.30, 2.04]	11.0	402	0.363
B. Incomes and households, 2015–2023							
Income per consumption unit (median)	0.08	(0.30)	[−0.42, 0.97]	[−0.44, 0.92]	16.0	695	0.310
Income per person	−0.09	(0.29)	[−0.60, 0.74]	–	16.0	695	0.318
Income per household	0.89	(0.37)	[0.34, 2.14]	–	16.0	695	0.293
Household size (income atlas)	1.26	(0.38)	[0.69, 2.51]	[0.68, 2.76]	16.0	695	−0.022
Household size implied by income per household and per person	0.99	(0.25)	[0.60, 1.75]	–	16.0	695	–
Number of households (population minus household size)	−0.41	(0.39)	[−1.65, 0.19]	–	16.0	695	–
C. Affordability							
Insiders: rent of all leases minus income per consumption unit	0.18	(0.34)	[−0.76, 0.81]	–	16.0	695	−0.144
Entrants: implied entry rent minus income per consumption unit	0.75	(0.37)	[−0.06, 1.74]	[0.00, 1.68]	11.0	402	0.057
Entrants, rents and incomes both over 2015–2023	0.90	(0.48)	[−0.16, 2.30]	–	9.6	402	–
D. Quantities and falsification							
Population	0.82	(0.27)	[0.17, 1.34]	[0.08, 1.38]	21.0	695	0.065
Dwelling stock (cadastre)	−0.26	(0.23)	[−0.81, 0.18]	[−0.92, 0.24]	21.0	695	0.038
Rent of all leases, 2012–2015 (pre-period)	−0.09	(0.13)	[−0.31, 0.28]	–	20.9	694	−0.035

Notes: one observation per municipality; coefficients in log points per percentage point of inflow. Treatment: cumulative net foreign-born inflow from 1 January 2015 to 1 January 2025 (to 1 January 2024 for the outcomes measured to 2023: panel B, the insiders row and the last entrants row) relative to 2015 population, instrumented with the corresponding sum of leave-own-province-out shift-share predictions (33 origins, 2003 settlement). All regressions include province fixed effects and predetermined controls (2011 census: log population, renting, vacancy, tertiary education, share aged 65+; 2012: construction, industry and trade-hospitality shares of firms, firms per capita; coastal and FUA-core indicators), 2015 population weights and standard errors clustered by province with the CRV1 small-sample correction. Entry-gap outcomes are available for the municipalities above 20,000 inhabitants in the AEAT statistics. AR: Anderson-Rubin set; WCR-AR: wild cluster restricted bootstrap Anderson-Rubin set (Webb weights, 4,999 draws); F: cluster-robust first-stage F. [a] The entry gap is observed only in 2024; the implied entry rent measures the growth of entry rents since 2015 only if the 2015 gap is unrelated to the instrument (section 6).

but the two measures cover different periods and definitions and are not additive. Income per consumption unit shows no detectable response (0.08, [−0.42, 0.97]), nor does income per person (−0.09); the Anderson-Rubin set rules out increases in income per consumption unit larger than about one log point per point of inflow. Household size rises by 1.26 [0.69, 2.51] and income per household by 0.89 [0.34, 2.14]. Prediction P2 holds in the sense that income per household rises with household size while income per consumption unit does not. Two qualifications apply. The household size implied by income per household and per person rises by 0.99 [0.60, 1.75], less than the atlas measure, and the number of households, population minus household size, shows no detectable change (−0.41, [−1.65, 0.19]). And larger households may reflect the composition of arrivals, whose households are larger, as much as more people sharing each dwelling; these data do not separate the two.

The point estimates imply that the affordability of entrants deteriorates by 0.75 log points per point of inflow while that of insiders does not change (0.18), but neither is precisely estimated: the set for entrants is [−0.06, 1.74], and 0.90 [−0.16, 2.30] when rents and incomes are both measured over 2015–2023. The dwelling stock shows no detectable response (−0.26, [−0.81, 0.18]), and the falsification test passes: the inflow of 2015–2024 does not predict rent growth in 2012–2015 (−0.09, [−0.31, 0.28]).

7.2 Who adjusts

Prediction P4 concerns quantities. Table 3, panel A, uses moves between municipalities in 2021–2024, instrumented with the shift-share prediction for the same years. A foreign-born inflow of one point is not associated with a detectable change in the out-migration of Spanish nationals to other municipalities (-0.01, $[-0.40, 0.27]$, against a mean of 9.9 moves per 100 residents over the four years). It lowers their in-migration from other municipalities by 0.54 (-0.54, $[-2.00, -0.14]$), so that their net migration falls by 0.53 ($[-1.82, -0.19]$). Foreign nationals move more in both directions, partly because a larger foreign population generates more moves.

These patterns are what P4 predicts: fewer would-be entrants arrive, and incumbents do not leave in detectable numbers. They are also compatible with native avoidance of immigrant destinations for reasons unrelated to rents (Card 2001; Saiz and Wachter 2011), which the data cannot rule out, since moves are not linked to tenure or rents. Naturalised residents count as Spanish nationals, and turnover within municipalities, the other half of P4, is not observed outside Aragón.

Table 3 Margins of adjustment: residential flows, heterogeneity, turnover and renters

A. Moves between municipalities per 100 residents, 2021–2024, per point of foreign-born inflow (2SLS)

	2SLS	SE	AR 95 % set	F	Mean
Out-migration of Spanish nationals	-0.01	(0.13)	$[-0.40, 0.27]$	10.1	9.93
In-migration of Spanish nationals	-0.54	(0.31)	$[-2.00, -0.14]$	10.1	9.59
Net migration of Spanish nationals	-0.53	(0.27)	$[-1.82, -0.19]$	10.1	-0.34
Out-migration of foreign nationals	0.85	(0.22)	[0.55, 1.86]	10.1	4.09
In-migration of foreign nationals	0.60	(0.16)	[0.36, 1.28]	10.1	4.05

B. Heterogeneity: inflow and inflow × indicator (2SLS)

	Entry gap		Rent, all leases		Share
	Inflow	× indicator	Inflow	× indicator	
Coastal municipality	-0.05 (0.38)	0.59 (0.23)	0.21 (0.17)	0.02 (0.13)	46 %
More than 100,000 inhabitants	0.54 (0.45)	0.05 (0.48)	0.48 (0.27)	-0.45 (0.20)	57 %
Core city of a FUA	0.30 (0.51)	0.28 (0.42)	0.59 (0.29)	-0.42 (0.18)	75 %
Renting share in 2011 above mean	0.12 (0.50)	0.62 (0.54)	0.50 (0.23)	-0.29 (0.24)	47 %

C. Turnover, updates and renters

Annual exit rate of leases, Aragón register: baseline (range); Zaragoza	17.4 % (14.9–19.2); 14.9 %	cohorts 2012–18: 17.4 %; 2019–25: 17.4 %
Weight of new leases in the INE index: 2021; 2024	17.6 %; 12.3 %	median lease duration, Aragón: 3.8 years
Entry gap (INE): 2021; 2024; 2024 without update caps	2.3; 12.7; 6.7	mechanical share of widening due to caps $\leq$ 58 %
Renters' median rent-to-income ratio (EPF): 2016; 2024	27.6 %; 25.4 %	share above 40 %: 26 %; 18 %
Rent per m^2 premium of foreign-national renters: 2016–18; change to 2022–24	-3.0 %; 3.5 (2.0)	young natives: change 0.6 (2.6)

Notes: panel A, INE Estadística de Migraciones y Cambios de Residencia (tables 69743 and 69745), moves between Spanish municipalities in 2021–2024 per 100 residents in 2021, by nationality (naturalised residents count as Spanish nationals); treatment: foreign-born net inflow over the same years relative to 2021 population (points), instrumented with the shift-share prediction for 2021–2024; controls, fixed effects, weights and clusters as table 2. Panel B: 2SLS with the inflow and its interaction with the indicator, instrumented with the prediction and its interaction; same specification. Panel C: exit rate from $S(a) = p + (1-p)e^{-ha}$ fitted to the share of each deposit cohort (2012–2025) not yet returned at the date of download; the age of each cohort is measured from mid-year (range: from the start of the year, or to the start of 2026); cohort rates fix p at its pooled value. The update-cap accounting adds back to existing-lease growth the difference between average CPI inflation and the cap for the share of each year in which the cap applied, an upper bound on its mechanical effect because not all leases are indexed to the CPI. EPF: INE household budget survey, households renting at market price (23,274 household-years, 2016–2024); income is the midpoint of the reported monthly income interval, available in every year; premium from a weighted regression of log rent per m^2 on nationality and age of the reference person by period, log floor area and year, region and municipality-size fixed effects; robust standard errors in parentheses.

7.3 Supply constraints and the coast

Table 4 and figure 5 test prediction P3. A one-standard-deviation increase in the share of undevelopable land within 10 km raises the effect of the demand shock on the entry gap by 0.49 (p = 0.021) and on implied entry rents by 0.54 (p = 0.018). The interaction is similar for the sea (0.32, p = 0.023) and for steep slopes (0.75, p = 0.071), and survives the inclusion of the interaction between the inflow and tourist exposure (0.49, p = 0.004). The constraint does not change the response of the rent of all leases (0.03) or of the cadastral dwelling stock (0.02), which is close to zero everywhere.

Three caveats limit this evidence. The partial first-stage F of the interaction is modest (5.8). At the population-weighted mean of the constraint the effect on the entry gap is 0.12 (standard error 0.37), and its 95 % band excludes zero only above about 61 % of undevelopable land, in 66 municipalities with 13 % of the population of the sample. And the constraint is concentrated on the coast. Interacting the inflow with a coastal indicator alone gives 0.59 (standard error 0.23), with a response of −0.05 in inland municipalities. Including both interactions, the geographic one keeps its size (0.70, standard error 0.44, p = 0.12) while the coastal one vanishes (−0.59, 0.78). Geography and the coast cannot be separated with these data. P3 is supported only in the weaker sense that the reset is larger where land is scarce, which in Spain is mostly on the coast. Table 3, panel B, also shows that the response of the rent of all leases is smaller in large cities (−0.45, standard error 0.20) and in core cities (−0.42, 0.18). That is what lower turnover there would produce, as in Zaragoza relative to the rest of Aragón, but it is not a test of the turnover prediction.

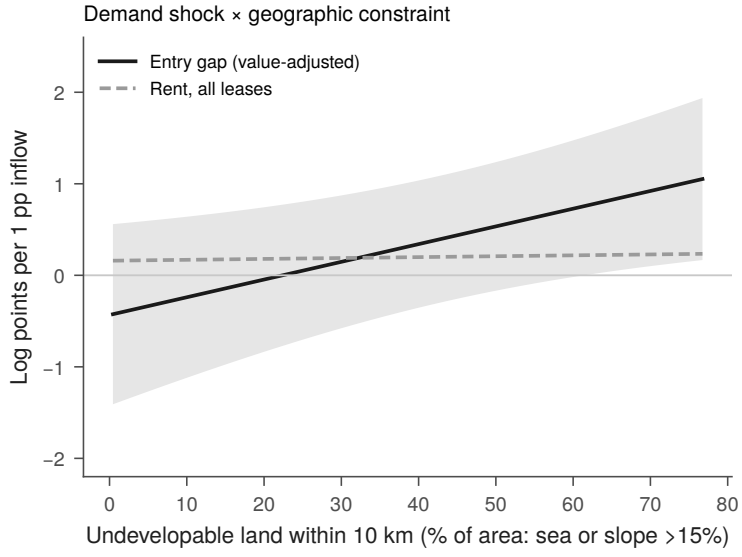


Fig. 5 Demand shocks and geographic constraints

Notes: marginal effect of a 1 pp inflow on the value-adjusted entry gap and on the rent of all leases (log points) as a function of undevelopable land within 10 km (5th to 95th percentile), from table 4; 95 % band for the entry gap.

Table 4 Demand shocks, geographic constraints and the coast

Outcome	Constraint	Inflow	Inflow × constraint	F (inflow;	Adding inflow × tourism		Adding inflow × coast	
		(at mean)		interaction)	× constraint	× tourism	× constraint	× coast
Entry gap (value-adjusted)	Undevelopable	0.12 (0.37)	0.49** (0.21)	10.3; 5.8	0.49*** (0.17)	−0.30** (0.14)	0.70 (0.44)	−0.59 (0.78)
	Sea	−0.02 (0.31)	0.32** (0.14)	25.7; 11.2	0.42*** (0.12)	−0.31** (0.13)	0.39 (0.27)	−0.29 (0.59)
	Slope >15 %	0.76 (0.50)	0.75* (0.41)	10.3; 9.3	0.58** (0.28)	−0.27 (0.20)	0.70 (0.48)	0.15 (0.37)
	Coast (indicator)	−0.05 (0.38)	0.59** (0.23)					
Implied entry rent	Undevelopable	0.34 (0.35)	0.54** (0.22)	10.3; 5.8	0.51** (0.20)	−0.25** (0.12)	0.70 (0.47)	−0.46 (0.79)
	Sea	0.19 (0.28)	0.35** (0.16)	25.7; 11.2	0.43*** (0.11)	−0.25** (0.10)	0.39 (0.31)	−0.17 (0.59)
	Slope >15 %	0.99 (0.53)	0.75* (0.40)	10.3; 9.3	0.56* (0.30)	−0.22 (0.19)	0.66 (0.45)	0.28 (0.35)
	Coast (indicator)	0.09 (0.36)	0.71** (0.28)					
Entrants' affordability	Undevelopable	0.40 (0.31)	0.38* (0.21)	10.3; 5.8	0.34 (0.23)	0.05 (0.13)		
	Sea	0.19 (0.29)	0.32* (0.18)	25.7; 11.2	0.29* (0.17)	0.05 (0.13)		
	Slope >15 %	0.84 (0.49)	0.50 (0.36)	10.3; 9.3	0.43 (0.38)	0.06 (0.16)		
Rent, all leases	Undevelopable	0.19 (0.14)	0.03 (0.11)	15.7; 8.0				
	Sea	0.22 (0.12)	0.00 (0.05)	50.4; 16.1				
	Slope >15 %	0.19 (0.18)	0.00 (0.15)	14.6; 12.2				
Dwelling stock	Undevelopable	−0.31 (0.25)	0.02 (0.09)	15.7; 8.0				
	Sea	−0.28 (0.29)	0.02 (0.07)	50.4; 16.1				
	Slope >15 %	−0.32 (0.24)	0.00 (0.13)	14.6; 12.2				

Notes: 2SLS with two endogenous regressors, the inflow and its interaction with a constraint standardised with population weights, instrumented with the shift-share prediction and its interaction; the inflow coefficient is the effect at the population-weighted mean of the constraint. Constraints: share of land within 10 km of the municipality centroid that is sea or has slope above 15 % (Copernicus DEM GLO-90), and each component. Same fixed effects, controls, weights and clustering as table 2. F: first-stage partial F statistics. The columns on tourism add the inflow interacted with the share of dwellings in tourist use (2020) and its instrument; the columns on the coast add the inflow interacted with the coastal indicator and its instrument (three endogenous regressors). The rows labelled Coast (indicator) interact the inflow with the coastal indicator alone. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

7.4 Tourism

The interaction between the inflow and tourist exposure in table 4 is negative (−0.30, $p = 0.028$): the same demand shock widens the entry gap less where many dwellings are tourist lets, as if part of that stock acted as a reserve for long-term letting. The collapse of tourism in 2020–21 offers a second, descriptive, look at this margin (table 5, panel A; figure 6). In Catalan municipalities where 3–8 % and more than 8 % of dwellings were tourist lets, the number of new long-term leases rose by 5.5 % and 6.7 % during the collapse relative to municipalities with little tourist letting. With the recovery, rents on new leases rose by about 3 % in the same groups (3.1 % and 3.1 %). In municipalities with 1–3 % of dwellings in tourist use, by contrast, new leases fell during the collapse (−4.4 %) and kept falling, and new-lease rents fell slightly (−2.6 %). The reallocation of leases is visible only above about 3 % of the stock.

These patterns are not causal estimates. In the annual register (274 municipalities), the number of new leases was already on different trends before 2019 in every exposure group ($p < 0.001$ for each), and so were rents in the most exposed group ($p < 0.001$). Tourist exposure is measured in August 2020, after the collapse began, because the INE statistics start then. A threshold regression (Hansen 2000) locates a change in the slope of the response at about 2.4 % of dwellings in tourist use (bootstrap p-value for no threshold < 0.002 for leases signed and 0.012 for rents). Given the pre-existing trends, the threshold describes where the patterns change, not a reallocation elasticity.

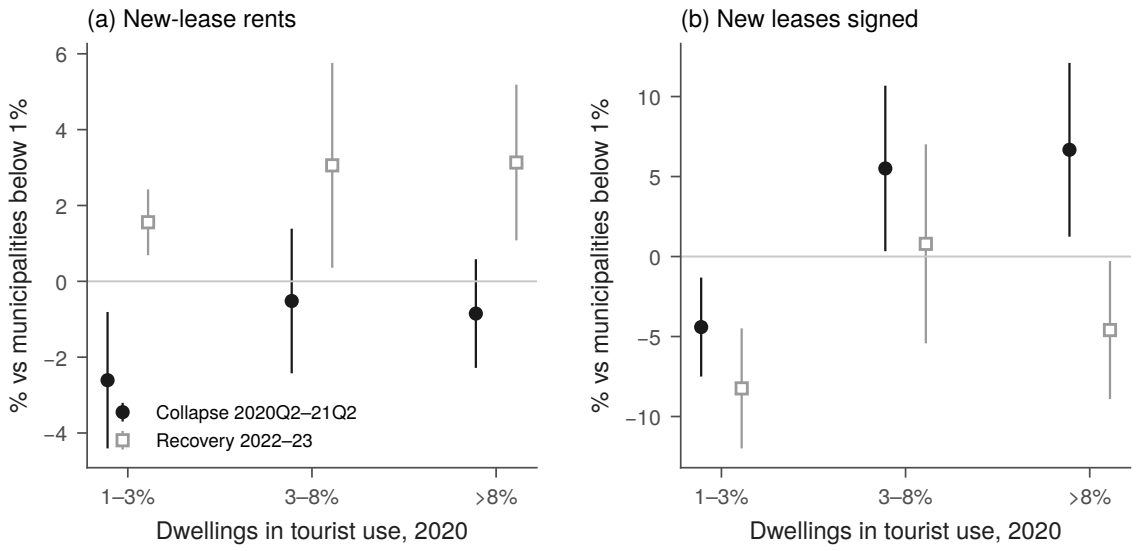


Fig. 6 Tourist dwellings and new leases during the collapse and recovery of tourism

Notes: coefficients and 95 % intervals of table 5, panel A, by share of dwellings in tourist use in August 2020. Descriptive: exposure groups were on different pre-2019 trends (table 5).

7.5 Regulation

Prediction P5 says that caps on updates widen the entry gap and that caps on entry rents narrow it. Between 2021 and 2024 the national entry gap widened in every year, from 2.3 to 12.7 log points. In 2022 existing leases rose by 2.1 log points while average inflation was 8.1. Suppose that existing leases had instead followed inflation for the share of each year in which the cap applied, adding 4.5 points in 2022 and 1.5 in 2023. The gap in 2024 would then have been 6.7 log points, so the caps account mechanically for at most 58 % of the widening (table 3, panel C). This is an upper bound on the mechanical effect, because not all leases are indexed to the CPI and updates use the inflation rate of the month before each anniversary rather than the annual average. Front-loading of entry rents in anticipation of capped updates would add to the effect of the caps, but the caps were temporary and unexpected, and in 2024 average inflation (2.7) was below the 3 % cap. The rest of the widening reflects faster growth of entry rents, which in 2024 rose by 8.4 log points against 2.8 for existing leases. Across provinces, the growth of existing and of new leases was more correlated in 2022 (0.89) than in 2021 (0.76): the cap held existing-lease growth down without severing its link to local markets.

The Catalan cap on new leases in stressed areas, evaluated in the companion analysis with stacked difference-in-differences, lowered new-lease rents by about 4.4 % relative to municipalities not yet designated (5.0 % relative to never-designated ones). It lowered the number of new leases by about 18 % in the first wave, although leases were already declining in designated municipalities before the cap (19 % after removing pre-period trends). These magnitudes are in line with evaluations of the 2020 Catalan law and of the 2024 cap (Jofre-Monseny et al. 2023; Monras and Montalvo 2023; Pérez García 2026). Table 5, panel B, shows where part of those leases went. After the cap, the share of seasonal leases

among new leases rose by 1.8 percentage points in the first wave, seasonal leases rose by about 26 %, and regular leases fell by 0.14 log points (standard error 0.03) in the 2023–2025 window of the seasonal register. In the whole sample, the share of seasonal leases rose from 6.6 % in 2023 to 11.1 % in 2025. The register of seasonal leases begins in 2023, so the pre-period is short (figure 7).

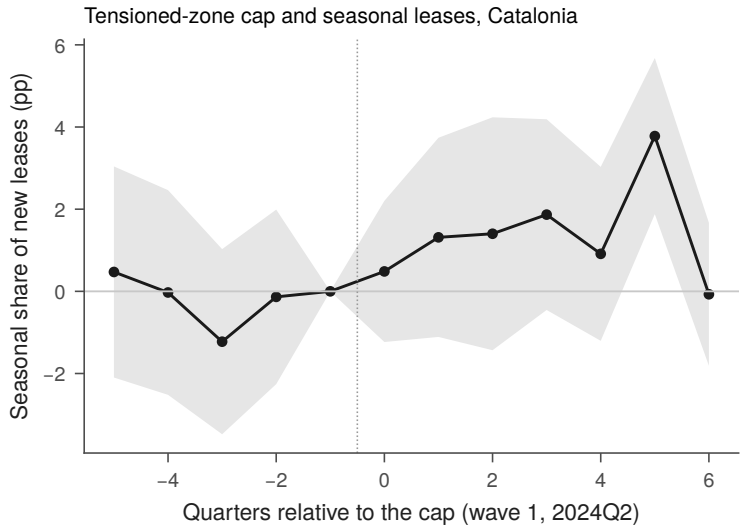


Fig. 7 Rent cap in stressed areas and seasonal leases

Notes: event study of the seasonal share of new leases in the first wave of designated municipalities relative to never-designated ones, 2023Q1–2025Q4; 95 % band; clusters by municipality.

7.6 Tenure, spillovers and renters

If the rise in mortgage rates from 2022 had pushed would-be buyers into renting, rents should have risen more where housing was most expensive relative to incomes. Table 5, panel C, shows no break after 2022 in the rent of all leases along that gradient and no association between price-to-income and the entry gap in 2024. But the gradient was already moving before 2021 (coefficients of 0.82 in 2019, p = 0.001, and 0.21 in 2020), so the design cannot isolate a tenure channel. Panel D studies spillovers within functional urban areas. The demand shock of the core city raises the dwelling stock of its peripheral municipalities (0.37, $p = 0.040$) but has small and imprecise effects on their rents and entry gaps. Moves point the same way: the core's shock raises the in-migration of foreign nationals to the periphery (0.29 per point, $p = 0.001$) but not detectably that of Spanish nationals (0.20, standard error 0.25).

The household budget survey does not identify recent movers, but households headed by a foreign national are likelier to have signed recently. Conditional on region, municipality size and floor area, their rent per m^2 was 3.0 % lower than that of other renters in 2016–2018 and rose by 3.5 points (standard error 2.0, $p = 0.08$) by 2022–2024, while that of young Spanish-headed households did not change (0.6, 2.6). The survey is consistent with an entry premium, but with these proxies the evidence is weak.

Table 5 Tourism, regulation, tenure and spillovers

A. Tourist dwellings and new leases, Catalonia (% relative to municipalities with <1 % of dwellings in tourist use); descriptive

Exposure in 2020	New-lease rents			New leases signed		
	Collapse	Recovery	2024–25	Collapse	Recovery	2024–25
1–3 %	−2.6***	1.6***	−0.9	−4.4***	−8.2***	−12.4***
3–8 %	−0.5	3.1**	3.3***	5.5**	0.8	9.8
>8 %	−0.8	3.1***	2.3*	6.7**	−4.6**	8.8**
Pre-trend test, annual register 2014–2018 (p)	0.080; 0.035; <0.001			<0.001; <0.001; <0.001		

B. Rent cap in stressed (tensioned) market areas and seasonal leases, Catalonia, 2023Q1–2025Q4 (DiD)

	Seasonal share (pp)	log seasonal leases	log regular leases
Wave 1 (2024Q2)	1.82*** (0.39)	0.26*** (0.04)	−0.14*** (0.03)
Wave 2 (2024Q4)	0.94** (0.40)	0.09*** (0.03)	−0.08*** (0.03)

C. Tenure: price-to-income in 2021 (s.d.) × year, rent of all leases (reference 2021)

	2019	2020	2022	2023	2024	Entry gap 2024
Coefficient	0.82***	0.21**	0.06	0.07	−0.01	−0.01

D. Spillovers: shift-share shock of the core city on peripheral municipalities of the same FUA (reduced form)

	Rent, all leases	Entry gap	Population	Inflow	Dwellings
Own shock	−0.04	0.13	−0.32**	−0.00	−0.20*
Core shock	0.13	0.29	0.39	0.03	0.37**

Notes: panel A, 274 Catalan municipalities with new leases in every quarter of 2019; municipality and quarter fixed effects and exposure-bin × calendar-quarter terms; collapse 2020Q2–2021Q2, recovery 2022–2023; population weights; clusters by municipality. Exposure: dwellings in tourist use in August 2020 (INE experimental statistics, which start that month). Pre-trend test: annual register, 274 municipalities, municipality and year fixed effects and exposure-bin × year terms (reference 2019), Wald test that the 2014–2018 terms of each bin are zero, clusters by municipality. Panel B: municipality and quarter fixed effects and tourist exposure × calendar quarter; clusters by municipality; the seasonal-lease register starts in 2023. Panel C: IPVA municipalities with an appraised value (MIVAU, municipalities above 25,000 inhabitants); municipality and province × year fixed effects and predetermined characteristics × year; coefficients ×100. Panel D: 217 peripheral municipalities in 48 functional urban areas; province fixed effects and controls; clusters by FUA. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

8 Robustness and limits of identification

Origins and controls. Table 6 collects the identification checks and table A1 decomposes the instrument. Colombia, Peru and Romania carry 84 % of the Rotemberg weight, but their origin-specific estimates of the effect on the entry gap are small (0.06, 0.15 and 0.46). Together they contribute 0.15 of the 0.57 baseline; the rest comes from the other origins, several of them with small weights and large origin-specific estimates. Using the components of the instrument for the three origins and for the rest as four separate instruments, two-step efficient GMM gives 0.21 (standard error 0.11) for the entry gap, 0.45 (0.11) for implied entry rents and 0.25 (0.06) for the rent of all leases. The Hansen tests do not reject the over-identifying restrictions at 5 % ($p = 0.12$, 0.34 and 0.08), but they have limited power, and GMM standard errors with few clusters are likely to be too small. Controlling for the 2003 foreign-born share (Borusyak et al. 2022) or for the 2003–2014 foreign-born inflow, which captures the persistence of settlement (Jaeger et al. 2018), leaves the baseline estimates almost unchanged (0.64 [0.24, 3.64] and 0.59 [0.13, 1.76] for the gap). Dropping any of the five origins with the largest Rotemberg weights leaves the effect on the gap between 0.56 and 0.96 and on implied entry rents between 0.87 and 1.02. The 2003 shares of Colombia and Peru are unrelated to rent growth in 2012–2015 conditional on the controls ($t =$

0.5 and −0.9), but the Romanian share is associated with slower pre-period growth ($t = -4.4$), and the three shares are related to other predetermined characteristics (table A1, panel B). A placebo by origin points in the expected direction but has little power. Inflows from high-income European origins, which more often buy than rent, do not raise the entry gap (−0.82, standard error 0.58), while the rest do (0.37, 0.24), but the first stage for the former is weak ($F = 1.3$).

The conclusion from these checks is that entry rents respond more than the rent of all leases under every identifying assumption considered, while the size of the response of the gap ranges from 0.21 to 0.57 log points per point of inflow, depending on how the origins are weighted.

Shock-based inference. Recentring the instrument by permuting the 33 origin shocks across origins removes its first stage ($F = 0.08$), because the realised shocks are dominated by a few origins with small 2003 stocks. Permuting within continents keeps a first stage ($F = 12.3$) but yields unbounded randomisation sets, and exposure-robust sets (Adão et al. 2019) are unbounded too. The long-difference estimates therefore rest on the exogeneity of settlement shares (Goldsmith-Pinkham et al. 2020). Wild cluster restricted bootstrap sets are close to the analytic Anderson-Rubin sets (table 2).

Weights and influential provinces. With square-root population weights the effect on the gap is 0.73 [0.00, 4.06]; without the provinces of Madrid and Barcelona it is 0.53 [−0.07, 3.06], and without Málaga 0.43 [0.00, 1.41]. Leaving out one province at a time gives estimates between 0.43 and 0.75, and the Anderson-Rubin set includes zero only without Málaga. The sets for implied entry rents exclude zero in every case. Without population weights the first stage is weak ($F = 2.3$), so the estimates describe the larger municipalities that hold most of the population.

The gap in changes and the timing of inflows. Table 7, panel A, uses the provincial INE indices, whose ratio in 2024 measures the change in the gap since 2015. The provincial inflow raises that change by 0.13 (standard error 0.22) with population weights, 0.39 (0.27) controlling for log population and 0.30 (0.32) unweighted. These estimates are positive but imprecise and smaller than the municipal estimate for the 2024 level. The change in the gap between 2021 and 2024 responds by 0.33 (0.14), $p = 0.027$. Panel B separates the inflows of 2015–2019 and 2020–2024. Equation (4) implies that the 2024 gap should respond mainly to recent inflows and the rent of all leases mainly to earlier ones. The point estimates go the other way: for the gap, 1.70 (1.06) for earlier and 0.09 (0.77) for recent inflows; for the rent of all leases, −0.14 (0.38) and 0.37 (0.26). But the two instruments have a correlation of 0.94, and the windows cannot be separated with useful precision. The timing implied by the model is neither confirmed nor rejected.

Annual panel. Panel C uses the companion paper's annual panel, in which origin shocks vary by year and randomisation inference is informative, merged with annual rents at signature in Catalonia, Aragón and the Comunitat Valenciana (2,097 municipality-years, 221 municipalities), without trimming. An

Table 6 Robustness of the long-difference estimates: identification, inference, weights and provinces

	Entry gap (value-adjusted)	Implied entry rent	Rent, all leases	First-stage F (gap)
A. Baseline (table 2)				
2SLS; AR set	0.57 (0.30)	0.83 (0.37)	0.22 (0.16)	11.0
	[0.11, 1.74]	[0.25, 2.21]	[−0.09, 0.62]	
WCR-AR set	[0.10, 1.54]	[0.30, 2.04]	[−0.06, 0.64]	
B. Identification				
Control: 2003 foreign-born share (sum of exposure shares)	0.64 (0.31)	0.84 (0.41)	0.18 (0.14)	5.6
	[0.24, 3.64]	[0.30, 4.80]	[−0.04, 0.91]	
Control: foreign-born inflow 2003–2014	0.59 (0.31)	0.85 (0.39)	0.22 (0.16)	11.6
	[0.13, 1.76]	[0.26, 2.30]	[−0.07, 0.62]	
Four instruments: Colombia, Peru, Romania, rest (two-step GMM)	0.21 (0.11)	0.45 (0.11)	0.25 (0.06)	
	Hansen J $p = 0.12$	Hansen J $p = 0.34$	Hansen J $p = 0.08$	
Single instrument: Colombia	0.06 (0.20)	0.59 (0.29)	0.47 (0.19)	14.7
Single instrument: Peru	0.15 (0.13)	0.27 (0.25)	0.04 (0.17)	17.0
Single instrument: Romania	0.46 (0.23)	0.67 (0.26)	0.24 (0.13)	4.4
Dropping one of the five origins with largest Rotemberg weights (range)	0.56 to 0.96	0.87 to 1.02	0.04 to 0.27	
C. Weights and influential provinces				
Square-root population weights	0.73 (0.52)	0.92 (0.58)	0.20 (0.22)	6.7
	[0.00, 4.06]	[0.12, 4.69]	[−0.18, 0.81]	
Without the provinces of Madrid and Barcelona	0.53 (0.41)	0.81 (0.51)	0.24 (0.20)	6.5
	[−0.07, 3.06]	[0.06, 3.98]	[−0.13, 0.81]	
Without the province of Málaga	0.43 (0.28)	0.68 (0.32)	0.23 (0.17)	12.1
	[0.00, 1.41]	[0.15, 1.78]	[−0.09, 0.62]	
Leaving out one province at a time (range)	0.43 to 0.75			

Notes: 2SLS coefficients (cluster-robust standard errors) and, below, Anderson-Rubin 95 % sets, for the specification of table 2; F refers to the entry-gap sample (402 municipalities; the rent of all leases uses 695). Panel B: the 2003 foreign-born share scaled as the instrument (Borusyak et al. 2022); the 2003–2014 foreign-born inflow relative to 2003 population (Jaeger et al. 2018); two-step efficient GMM with the components of the instrument for the three origins with the largest Rotemberg weights and the sum of the other 30 as separate instruments, with the Hansen test of over-identifying restrictions; the just-identified estimates with each origin component alone (the F column reports the first stage in the gap sample); and the range of estimates when one of the five origins with the largest Rotemberg weights is dropped from the instrument. Panel C: the range across the 42 leave-one-province-out estimates of the entry gap; its Anderson-Rubin set includes zero only when Málaga is excluded.

inflow of 1 % raises new-lease rents by 0.33 (standard error 0.75; Anderson-Rubin set [−2.41, 1.86]) and the rent of all leases by 0.01, so the entry gap widens by 0.32 (0.74). The estimates are imprecise ($F = 7.9$; randomisation p-values 0.13 for entry rents and 0.17 for the gap), and next year's predicted inflow is negatively related to this year's entry rents (−1.17, standard error 0.79). The panel does not contradict the direction of the cross-sectional estimates, but it is uninformative about their size.

Measurement. Table A2, panel A, shows that the effect on the gap does not depend on the value adjustment: 0.71 for the raw gap, 0.64 and 0.61 with value elasticities of 0.5 and 0.7, and 0.57 with an elasticity of one. The demand shock does not detectably change the relative value (0.15, [−0.40, 0.58]) or floor area (0.22, [−0.51, 0.57]) of the dwellings let on new leases.

Instrument and registers. Table 7, panel D, reports alternative instruments and registers. With the companion paper's instrument, which sums annual predictions with centred timing over its 699 municipalities, the effect on the rent of all leases is larger (0.41) and the effect on the value-adjusted entry gap is 0.43 [−0.03, 1.27]; the raw gap responds by 0.57 [0.15, 1.40]. In the Basque register, which measures rents per m^2, the demand shock raises entry rents by 0.85 [−1.42, 2.26] and worsens entrants' affordability by 1.23 [0.30, 3.10]. The Catalan register shows why adjusting for the dwelling matters.

Its mean rent per contract falls with the demand shock (-0.55, $[-1.25, -0.04]$), consistent with arrivals shifting new leases towards cheaper and smaller dwellings; the tax measure, which adjusts for cadastral value, does not show such a shift.

Table 7 The entry gap in changes, timing, annual panel and other registers

A. Provinces: INE indices of new and existing leases (base 2015 = 100), 2SLS on the provincial inflow				
	Population weights	Adding log population	Unweighted	First-stage F
ln(new/existing) in 2024 = change since 2015	0.13 (0.22)	0.39 (0.27)	0.30 (0.32)	5.6
Change in ln(new/existing), 2021–2024	0.33** (0.14)	0.47** (0.18)	0.40 (0.25)	5.6
B. Earlier and recent inflows (two endogenous regressors)				
	Inflow 2015–2019	Inflow 2020–2024	Correlation of instruments	N
Entry gap (value-adjusted), 2024	1.70 (1.06)	0.09 (0.77)	0.94	402
Implied entry rent	1.52 (0.97)	0.54 (0.57)	0.94	402
Rent, all leases, 2015–2024	−0.14 (0.38)	0.37 (0.26)	0.94	695
C. Annual panel with year-varying shocks: Catalonia, Aragón and Comunitat Valenciana, 2012–2024				
	2SLS (SE)	AR 95 % set	Randomisation set^r	F; N
New-lease rent, annual growth	0.33 (0.75)	[−2.41, 1.86]	[−0.38, 1.18]	7.9; 2,097
Rent of all leases, annual growth	0.01 (0.11)	[−0.43, 0.22]	[−0.18, 0.24]	7.9; 2,097
Change in the entry gap	0.32 (0.74)	[−2.24, 1.90]	[−0.46, 1.15]	7.9; 2,097
D. Instrument, weights and registers				
	2SLS (SE)	AR 95 % set	F	N
Companion-paper instrument (annual, centred timing), rent of all leases	0.41 (0.16)	[0.03, 0.75]	17.9	695
Companion-paper instrument, entry gap (value-adjusted)	0.43 (0.29)	[−0.03, 1.27]	20.0	402
Companion-paper instrument, entry gap (raw)	0.57 (0.27)	[0.15, 1.40]	20.0	402
Unweighted, rent of all leases	0.25 (0.37)	[−0.36, 1.79]	8.7	695
Unweighted, entry gap (value-adjusted)	1.09 (1.13)	[−∞, ∞]	2.3	402
Basque Country, new-lease rent per m^2, 2016–2023	0.85 (0.75)	[−1.42, 2.26]	11.5	67
Basque Country, entrants' affordability (per m^2)	1.23 (0.59)	[0.30, 3.10]	17.2	63
Catalonia, mean new-lease rent (not adjusted for size or value), 2015–2023	−0.55 (0.30)	[−1.25, −0.04]	61.4	288
Catalonia, income per consumption unit, 2015–2023	0.62 (0.44)	[−0.42, 1.38]	61.4	288

Notes: panel A, the 46 provinces of the common tax regime plus Ceuta and Melilla (INE table 59005, both indices with base 2015 = 100); treatment and instrument are the population-weighted provincial means of the municipal inflow and instrument of table 2; heteroskedasticity-robust standard errors. Panel B: inflows and instruments summed over 2015–2019 and 2020–2024, specification of table 2. Panel C: the panel of IPVA municipalities of the companion paper (annual inflow and instrument with centred timing, province × year fixed effects, 2003 foreign share and predetermined covariates × year, population weights) merged with annual new-lease rents (Catalonia: mean rent; Aragón: street-adjusted index; Comunitat Valenciana: median deposit; municipalities with at least 30 leases in the year); clusters by municipality; no trimming. ^{r} Randomisation-inference set: origin shock paths permuted across origins (2,000 draws), recentred instrument, cluster-robust t statistic. Panel D: companion-paper instrument, sum of annual centred shift-share predictions over 2016–2024 for its 699 IPVA municipalities (Fernández-Aguilera 2026). Basque Country and Catalonia: long-difference instrument of this paper; controls are the 2011 log population, renting and vacancy shares; province fixed effects (Basque Country: three provinces, clusters by municipality). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

9 Discussion

What the evidence supports. Rents at signature have grown faster than the rents of sitting tenants in Spain since 2021 and, in most registers, faster than incomes since 2015. Leases turn over slowly enough for the index of all leases to lag the market by years. Local demand shocks raise entry rents more than the rents of all leases, while incomes per consumption unit show no detectable response and households grow larger. Fewer Spanish nationals move into municipalities that receive larger inflows, while out-migration shows no detectable change. These findings are consistent with predictions P1, P2 and P4. The evidence on P3 and P5 is weaker, and the timing of the response implied by the model is not borne out by the point estimates.

What it does not establish. The size of the response of the entry gap is uncertain, between 0.21 and

0.57 log points per point of inflow, and the provincial indices that measure the gap in changes give smaller and imprecise estimates. The response of entry-rent growth since 2015 is identified only if the 2015 gap was unrelated to the instrument. The model's timing prediction is neither confirmed nor rejected. The affordability of entrants deteriorates in the point estimates, but not significantly. The divergence between entry rents and incomes depends on the base year and is reversed in the Basque Country. Amplification by geography cannot be separated from the coast, the tourism results are descriptive, and the tenure channel cannot be isolated. The prediction that distinguishes the model most sharply, a larger response of the gap where turnover is lower, is not tested.

Alternative interpretations. Four alternatives deserve mention. First, new leases might be on worse dwellings; but the value adjustment moves new leases further above old ones, and the shock does not change the relative value or size of new leases. Second, natives may avoid immigrant destinations for reasons unrelated to rents (Card 2001; Saiz and Wachter 2011), which would also lower their in-migration. Third, part of the growth of entry rents after 2015 is a cyclical recovery, which the base-year comparisons make visible; with sticky rents in force, the stock-flow mechanism implies that rents at signature also fell further in the downturn. Fourth, the tax measure covers individual landlords and may be affected by changes in the tax treatment of new leases. None of these alternatives explains the core fact: the rent of all leases grew less than incomes in every municipality covered, while rents at signature have grown faster than the rents in force in every province since 2021.

Welfare. A larger entry gap transfers resources from movers, young households and new arrivals to the landlords who relet and to sitting tenants. It also discourages mobility: households that would gain from moving for a job stay put, and those who need to move pay the reset (Glaeser and Luttmer 2003). Larger households are an adjustment margin, as the companion paper documents, and not by themselves a welfare loss. These welfare implications follow from the model and the facts; the paper does not estimate them.

10 Conclusion

The rise in Spanish rents has fallen mainly on households that enter the rental market. Official statistics show the rents of sitting tenants growing more slowly than incomes everywhere. Tax data and deposit registers show rents at signature growing faster than incomes in most markets, and a value-adjusted gap of 15.9 log points between new leases and leases in force in 2024. The two facts are consistent. In a market of long, indexed leases, of which fewer than one in five turn over each year, the average rent of all leases absorbs only a fraction of what entrants pay, and with a lag of several years.

The paper explains this divergence with a stock-flow model in which local demand shocks are absorbed by rent resets, and tests its predictions with an immigration-driven demand shock. Entry rents

respond more than the rents of all leases. Incomes per consumption unit show no detectable response and households grow larger. Fewer households move in, while incumbents do not leave in detectable numbers. The reset appears larger where land is scarce, and regulation that protects incumbents widens the gap while caps at signature shift lets towards unregulated contracts. The evidence on the size of these responses is less firm than the evidence on their direction, because it relies on the 2003 settlement of a few origin groups and on a single year of the entry gap.

Three implications for policy follow, with different degrees of support. The first concerns measurement and is the best supported. An index of all leases is the right deflator for what sitting tenants pay but a poor indicator of access to housing. Statistics of rents at signature, adjusted for the value and location of dwellings, exist in four regions and in the tax data for one year; publishing them annually for all of Spain would show where the problem is and how fast it grows. The second concerns supply and is an implication of the model rather than a result. Because demand shocks are absorbed at entry, measures that add dwellings to the flow available to entrants, through construction where land allows it and through the conversion of underused dwellings where it does not, would act where the problem arises. No such measure is evaluated here. The third concerns regulation. Caps on updates and longer minimum durations protect insiders and, by the accounting identity of the model, widen the gap. Caps at signature narrow it in the regulated segment, but reduce the number of regulated leases and shift lets to segments the cap does not cover. A policy mix that protects tenants without closing the market to newcomers would have to combine regulation with an expansion of the entry flow.

Three extensions would sharpen these conclusions. Linking successive deposits on the same dwelling would measure the reset directly, remove the need for value adjustment and give the change in the gap rather than its level. Observing turnover by municipality would allow a test of the model's most distinctive prediction, that the gap responds more where leases last longer. And observing the incomes of the households who sign new leases would measure the affordability of entrants rather than infer it from local averages. All three require administrative data that the registers already hold.

Declarations

Funding No funding was received for this work.

Competing interests The author declares no competing interests.

Data and code availability All data are public and are downloaded by the replication code from INE, the tax agency (AEAT), the Generalitat de Catalunya, the Gobierno de Aragón, the Gobierno Vasco, the Generalitat Valenciana, the Ministry of Housing, the European Space Agency (Copernicus DEM) and Eurostat. The code that builds the instrument is shared with the companion paper and is included in the replication package.

Use of generative AI Claude (Anthropic) was used to retrieve the data and to assist with the code, the drafting and the English translation. The author verified all results and text and is fully responsible for the article.

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Appendix: additional tables

Table A1 Shift-share diagnostics: Rotemberg weights, origin-specific estimates and balance

A. Rotemberg decomposition of the entry-gap estimate and just-identified estimates by origin						
Origin	Weight α_o	$\hat{\beta}_o$ (gap)	Entry gap	Implied entry rent	Rent, all leases	First-stage F
Colombia	0.438	0.06	0.06 (0.20)	0.59 (0.29)	0.47 (0.19)	14.7
Peru	0.209	0.15	0.15 (0.13)	0.27 (0.25)	0.04 (0.17)	17.0
Romania	0.192	0.46	0.46 (0.23)	0.67 (0.26)	0.24 (0.13)	4.4
Ecuador	0.052	−0.11				
Rest of America	0.051	0.62				
Other 28 origins (net sum)	0.059					
All origins except Colombia, Peru and Romania (one instrument)			2.62 (3.60)	2.40 (3.17)	−0.13 (0.84)	0.7
Sum of negative weights	−0.205					

B. Balance: standardised coefficient (t statistic) of the 2003 origin share on predetermined characteristics, conditional on controls						
	Rent growth 2012–15	Renting 2011	Vacancy 2011	Tertiary 2011	Aged 65+ 2011	Coastal
Colombia	0.02 (0.5)	0.21 (3.1)	0.01 (0.2)	0.01 (0.2)	−0.19 (−3.3)	−0.03 (−0.6)
Peru	−0.06 (−0.9)	0.17 (2.5)	−0.06 (−1.7)	0.19 (4.2)	0.17 (5.3)	0.13 (1.8)
Romania	−0.08 (−4.4)	0.14 (3.2)	−0.05 (−2.0)	−0.01 (−0.2)	−0.16 (−3.2)	0.06 (3.3)

Notes: panel A, Rotemberg weights α_o and origin-specific estimates $\hat{\beta}_o$ of the entry-gap coefficient (Goldsmith-Pinkham et al. 2020), $\hat{\beta} = \sum_o \alpha_o \hat{\beta}_o$; just-identified 2SLS with the component of the instrument of each origin (2003 share × leave-out national growth) as the only instrument, specification of table 2; the F column refers to the entry-gap sample. Panel B: regression of each characteristic (standardised) on the 2003 share of the origin, with the controls other than the characteristic, province fixed effects and population weights; clusters by province.

Table A2 Measurement: value adjustment, composition, base year and the Basque register

A. Entry gap with value elasticity θ: $G^\theta = \ln(rent_{new}/rent_{all}) - \theta \ln(value_{new}/value_{all})$				
	Mean gap (log points)	2SLS (SE)	AR 95 % set	N
θ = 0 (raw gap)	13.9	0.71 (0.27)	[0.31, 1.72]	402
θ = 0.5	14.9	0.64 (0.27)	[0.25, 1.69]	402
θ = 0.7	15.3	0.61 (0.28)	[0.20, 1.70]	402
θ = 1 (baseline)	15.9	0.57 (0.30)	[0.11, 1.74]	402
Composition: $\ln(\text{value}_{new}/\text{value}_{all})$		0.15 (0.20)	[−0.40, 0.58]	402
Composition: $\ln(\text{m}^2_{new}/\text{m}^2_{all})$		0.22 (0.21)	[−0.51, 0.57]	402

B. Base year of the entry facts: growth to 2023 (log points, population-weighted)				
	New-lease rents	Income per consumption unit	Difference (share > 0)	N
Catalonia, from 2015	39.0	27.3	11.7 (77 %)	288
Catalonia, from 2016	33.1	25.0	8.1 (74 %)	298
Catalonia, from 2018	19.5	17.8	1.7 (55 %)	311
Aragón, from 2015	36.1	26.5	9.5 (77 %)	43
Aragón, from 2016	32.8	25.8	6.9 (74 %)	43
Aragón, from 2018	21.8	17.8	4.0 (81 %)	43
Catalonia, mean new-lease rent, 2008–2015; 2015–2023		−19.7; 40.5		

C. Basque Country: entry rent per m^2 minus income, 2016–2023, on the inflow (OLS, within province)		
Inflow 2015–2023	0.65 (0.28)	64

Notes: panel A, AEAT 2024; $\theta = 1$ is the value-adjusted gap of the main text; composition rows regress the log ratio of the mean cadastral value (or floor area) of new to all leases on the inflow; specification of table 2. Panel B: Catalonia, mean rent of new leases in municipalities with at least 30 leases in the base year; Aragón, street-adjusted index; income from the INE household income atlas. The last row uses all municipalities in the Catalan register, weighted by leases, and is not adjusted for composition. Panel C: municipalities with rents per m^2 in the Basque register, province fixed effects, population weights, robust standard errors.